MILITARY SPECIFICATION
ACTUATOR, ROTARY 1000-41
ASSEMBLY

1. SCOPE

1.1 This specification covers Actuator, Rotary 1000-41 Assembly, loaded with cartridge 5000-1, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MS-28741	- Hose Assembly - Detachable End Fitting, Medium Pressure

DRAWINGS

Munitions Command

C9218443	- Carton, Packing, Ammunition, for Rotary Actuator, 1000 or 1925 Series
C9218444	- Box, Packing Ammunition, for Rotary Actuator, 1000 or 1925 Series

Talley Industries (Code Ident. No. 12116)

1000-41	- Rotary Actuator Assembly

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2.1 (Cont'd)

(Copies of specifications, standards and drawings required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Breakaway torque.-All threaded joints of the assembled Actuator shall withstand a breakaway torque application of 200 inch pounds without loosening or deformation.

3.3 Residual magnetism.-The Actuator shall not deflect the indicator of the approved compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Initial lock security.-The lock pin shall secure the piston component of the ball screw assembly so that the strap drum component is not free to rotate.

3.5 Functional.

3.5.1 The actuator shall not fail to fire.

3.5.2 The actuator shall function without mechanical failure (See 6.2).

3.5.3 Initiation.-The actuator shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, Size 4, MS28741, at 70° ± 5°F.

3.5.4 Ballistics.-The Actuator loaded with an accepted cartridge and temperature conditioned, when fired, shall separate and eject the load specified in Table I from the approved simulated aircraft seat within an operating time of .150 seconds minimum and .500 seconds maximum and function within the performance limits specified in Table I.

TABLE I

Temperature	Pressure (psi)	Acceleration (g)	Ejected Load
-65°F to -70°F	*	*	400 lbs.
65°F to 75°F	2000 to 4800	0.85 to 3.0	400 lbs.
200°F to 205°F	*	*	100 lbs.

* Information only.

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3.6 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.-All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defects.

3.6.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might in injurious to personnel handling the item.

3.6.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.-A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of accepted cartridges.

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4.2.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All nonconforming material shall be rejected. Four power magnification may be used in performing visual examination for inspection of such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3
1801 - 10000	200	2	3	3	4

4.2.2.1 Cartridge Assembly, Drawing 5000-1, covering a detail of Drawing 1000-41.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Closure cap missing, damaged or improperly sealed	Visual
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Foreign matter	Visual
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Protective finish, missing or inadequate	Visual

4.2.2.2 Assembly, Drawing 1000-41 with End Closure Assembly (Drawing #1000-132) removed.

CRITICAL	
1. Lock pin loose	Manual/Visual
2. Ball screw and piston assembly loose	Manual

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4.2.2.2 (Cont'd)

Defect	Method of Inspection
CRITICAL	
3. Blowout plug loose	Manual/Visual
MAJOR	
101. Thread damaged	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Assembly Drawing 1000-41 with Breech Assembly (Drawing #1022) and Back Up Washer (Drawing #1017) removed.

CRITICAL	
1. Cartridge loose	Manual/Visual
2. Cartridge retainer loose	Manual/Visual
3. Foreign matter in gas port	Visual
MAJOR	
101. Thread damaged	Visual
MINOR	
None defined	

4.2.2.4 Assembly Drawing 1000-41.

CRITICAL	
1. Any part missing except as otherwise classified	Visual
2. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Length overall	Gage
102. Length of strap extended	Gage

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4.2.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR	
103. Shear studs loose	Manual/Visual
104. Cap screw loose	Manual/Visual
105. Threads damaged	Visual
MINOR	
201. Location labels incorrect	Visual
202. Shipping plug missing	Visual
203. Protective finish inadequate or defective	Visual

4.2.2.5 Packing in Fiber Box, Drawing C9218443.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Sealing strips missing, loose or improperly applied	Visual
202. Contents loose	Manual
203. Flaps loose or distorted	Visual
204. Cut or damaged through all impregnated layers	Visual
205. Marking, missing incorrect or illegible	Visual

4.2.2.6 Packing in Wooden Box, Drawing C9218444.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual

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4.2.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
202. Hardware missing, loose, broken or improperly engaged	Manual/Visual
203. Strapping missing, loose, broken or improperly assembled	Manual/Visual
204. Contents of box exposed	Visual
205. Marking missing, incorrect or illegible	Visual
206. Number of units incorrect	Visual
207. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection.-The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.-Each actuator shall be subjected to the breakaway torque application. Failure of any acturator to comply with the requirements of 3.2 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism.-Each actuator shall be subjected to the test for excessive residual magnetism. Failure of any actuator to comply with the requirements of 3.3 shall be cause for rejection of the actuator.

4.2.3.3 Initial lock security.-Each actuator shall be subjected to an initial lock test. Failure of any actuator to comply with the requirements of 3.4 shall be cause for rejection of the assembly.

4.2.3.4 Functional tests.-The following tests shall be performed by the Government.

4.2.3.4.1 Ballistics.-Twenty-one units shall be tested as specified in 4.3.4 to determine compliance with the requirements of 3.5.1 to 3.5.4. The units shall be temperature conditioned as provided in Table III and 4.3.4.1. Failure of any unit to comply with the requirements shall be cause for rejection of the lot.

TABLE III

Temperature	Sample Size
-65° to -70°F	7
65° to 75°F	7
200° to 205°F	7

NOTE: All variable data gained at -65°F and 200°F will be recorded and reported for information only.

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4.2.4 Inspection equipment.-The examination and tests shall be made using equipment approved by the Government.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.-A torque wrench, set at the torque requirements specified in 3.2 shall be used in conjunction with an adapter to determine the adequacy of the joints. The actuator shall be secured in a holding device and the breakaway torque applied to the end closure and breech assemblies.

4.3.2 Initial lock security.-The actuator shall be tested for initial lock security by securing the actuator body in a holding device and manually pulling on the strap assembly to assure that the webbing cannot be extended or retracted.

4.3.3 Residual magnetism.-The actuator shall be tested for excessive residual magnetism, in an area free of local magnetic effects, by placing the actuator six inches from and at the same height as the compass in a north-south horizontal position (breech end at south) with the compass set in an east-west heading as shown in Figure I. The actuator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly past the compass. This procedure shall be repeated for each 90 degree rotation of the actuator.

4.3.4 Functional tests.

4.3.4.1 Temperature conditioning.-Prior to conducting the ballistic test specified in 4.3.4.3, the actuator samples shall be subjected to the temperature condition specified in 3.5.4. A simulated actuator loaded with a dummy cartridge containing a temperature sensitive element shall be placed in the temperature conditioning chamber at the same time as the actuator samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.3.4.1.1 Upon completion of the required temperature conditioning, each actuator sample shall be ballistically tested within 5 minutes after removal from the conditioning chamber. Samples shall be reconditioned if this time is exceeded.

4.3.4.2 Method of testing.-The actuator tested under the conditions specified herein shall be mounted on an approved simulated aircraft seat in the proper position. Install a fixture with the load specified in Table I of 3.5.4 on the

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4.3.4.2 (Cont'd)

seat. The fixture shall be designed to cover at least 80% of the area comprising the bottom and back of the seat. Removable weights may be incorporated in the fixture so that the weight of the ejected mass can be adjusted from 100 to 400 pounds. Mount an accelerometer on the fixture to record acceleration of the ejected mass on a line which bisects the angle between the seat bottom and the seat back. A recording system shall be used to determine the amount of time required to eject the mass after actuation of the actuator under load conditions.

4.3.4.3 Ballistic test.-Utilize the test fixture specified in 4.3.4.2. Weights installed in the fixture shall be secured to prevent shifting. An M3A1 initiator or simulated method shall be used to actuate the actuator under test. Pressure sensitive devices shall be used to measure the pressure exerted to actuate the actuator except at -65° firings. Pressure sensitive devices shall also be used to measure the amount of ballistic gas pressure in the actuator during operation. A pressure port shall be installed in the actuator end closure assembly for recording the ballistic gas pressure. The pressure sensitive device for measuring the amount of pressure used to actuate the actuator shall be inserted into the system by means of a Tee coupling between the actuator and the actuating device except during -65° firings. The actuator sample shall be tested in accordance with the functional requirements of 3.5 within the time specified.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.-Packing and marking shall be in accordance with drawings C9218443 and C9218444.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:

Army - MU

Preparing activity:

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Project No. 1377-A377

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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